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END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering

Semester: III

Name of the Paper: Fluid and Particle Mechanics

Course Code: TPE-303

Time: Three Hours

MM: 100

Note:

- All questions are compulsory.
- Answer any two sub questions in each main question.
- Total marks for each main question is **twenty**
- Each Question carries **10 marks**

Q1		COs
a.	Explain the following with their SI Unit - Surface Tension - Froude number - Specific weight - Specific gravity - compressibility	CO1, CO2
b.	Derive an expression for the pressure in terms of surface tension for water droplet. Hence calculate the surface tension in a water droplet of 40mm diameter when the inside pressure is 2.5 N/m^2 .	CO1, CO2
c.	Define absolute, Gauge and vacuum pressure? Derive an expression for pressure variation in fluid at rest using the hydrostatic law?	CO3
Q2		
a.	Figure below shows a conical vessel having to which a U-tube manometer is connected. The reading of the manometer indicated in the figure pertains to the situation, when the vessel is empty. Find the reading of the manometer when the vessel is completely filled with the water.	CO3, CO4
b.	In a fluid, the velocities field is given by	CO1,

	$V = (6x + 4y)i + (4z + 6x^2)j + (4t - 6z)k$ Determine: i. The velocity component u, v, w at any point in the flow field ii. The speed at point (2,2,2) iii. The speed at time $t = 2s$ at point (0,0,2)	CO4
c.	A pipe (1) 500mm in diameter branches into two pipes (2 and 3) of diameter 300mm and 200mm respectively. If the average velocity in 500 mm diameter pipe is 6m/s find? i. Discharge through 450 mm diameter pipe ii. Velocity in 200mm diameter pipe if the average velocity in 300mm pipe is 2m/s,	CO1, CO5
Q3		
a.	Discuss the different types of forces acting on the fluid element in the fluid dynamics? Explain Euler's equation? Calculate: Water is flowing through a pipe of diameter 10 cm under a pressure of 30 N/cm ² and with mean velocity 2.5 m/s. Find the total energy per unit weight or the head of water at a cross section which is 5.5 m above the datum line?	CO1
b.	Derive an expression for the rate of flow or discharge (Q) over a <i>triangular notch or weir</i> mentioning angle of notch as ' θ ' Express the equation when angle of notch as [$\theta = 90^\circ$ and 60°]	CO1, CO2
c.	What are the different instruments of measurement of rate of flow of fluid through a pipe? Define Venturimeter and its part with the diagram and hence derive an expression for the actual discharge through the Venturimeter? <i>Specify the symbols used during derivation.</i>	CO2, CO5
Q4		
a.	What do you mean by viscous fluid of flow explain by showing the shear stress and velocity distribution diagram in the circular pipe? Hence Derive: i. Velocity distribution ii. Ratio of maximum velocity to average velocity (U_{max}/u_{avg}) iii. Pressure drop over a given length of pipe (L) i.e., <i>the equation of Hagen Poiseuille</i> .	CO1, CO5
b.	A pipe of length 1000 m and diameter 10 cm is laid at a slope of 1 in 200. An oil of sp. Gr. 0.9 and viscosity of 1.5poise is pumped up at the rate of 25 liters per second. Find i. head lost due to friction ii. the power required to pump the oil	CO4, CO5
c.	Define and classify the loss of energy in the pipes both major and minor energy losses? A fluid of kinematic viscosity of 0.5 stoke is flowing through a pipe of diameter 300mm at the rate of 300 liters per second. Find the head lost due to friction over a length of 50m of the pipe. Given that: 1stoke = 1 cm²/sec.	CO1, CO5

Q5		
a.	What is Pump? Categories the different types of pump? Discuss the construction of centrifugal pumps with neat and labeled diagram.	CO4, CO6
b.	Define the following for the motion of particle in fluid i. Stokes law ii. Drag coefficient iii. Terminal falling velocity iv. What is the terminal velocity of a spherical steel particle, 0.40 mm in diameter, settling in an oil of density 820 Kg/m^3 and viscosity of 10 mN/sec m^2? The density of steel is 7870 Kg/m^3.	CO4, CO5
c.	Discuss the following i. Packed beds ii. Fluidized beds iii. Screening iv. Effectiveness of screen?	CO2, CO6

